VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 6 | JEE - 2023 (Gen 1 & 2)

Date: 16/11/2021 **Maximum Marks: 300**

Timing: 04:00 PM - 09:00 PM **Duration: 3.0 Hrs**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

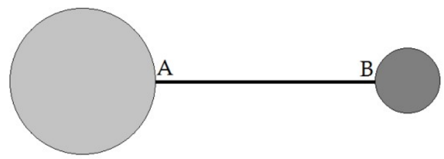
Candidate's Signature : _ _ _ _ _ _ _ _ _ _ **Invigilator's Signature** _ _ _ _ _ _ _

PART - I : PHYSICS 100 MARKS

SECTION-1

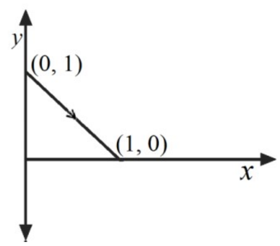
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Two uniform spheres of radius $2R$ and R, and mass M and $2M$ respectively, are attached to a uniform rod of length $8R$ and mass M at the points A and B such that the centres of both spheres are collinear with the rod. The distance of the centre of mass of the system from the point A is:



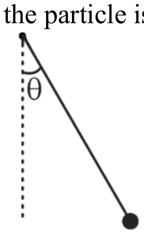
(A) $\frac{19R}{4}$ **(B)** $5R$ **(C)** $\frac{11R}{2}$ **(D)** $6R$

2. A particle moves from the point $(0, 1)$ to the point $(1, 0)$ along a straight line as shown. One of the forces acting on the particle was $\vec{F} = 10\left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)$. All quantities are in SI units. The work done (in Joules) on the particle by F is equal to:



(A) $5\sqrt{3}$ **(B)** 5 **(C)** $5\left(\sqrt{3}-1\right)$ **(D)** $5\left(1-\sqrt{3}\right)$

3. A small particle of mass m is tied to a massless thread and the other end of the thread is fixed at a point. The particle is released from rest with the string being tight and making an angle $\theta = 45^{o}$ with the vertical as shown. The tension in the thread when the particle is at the lowest point in its path is:



(A) $\left(3-\sqrt{2}\right)mg$ **(B)** $\left(\sqrt{2}-1\right)mg$ **(C)** $\left(2\sqrt{2}-1\right)mg$ **(D)** $\left(2-\sqrt{2}\right)mg$

4. A particle collides obliquely with a fixed smooth wall. Consider the following statements:

(I) the direction of the change in momentum of the particle during the collision is perpendicular to the wall

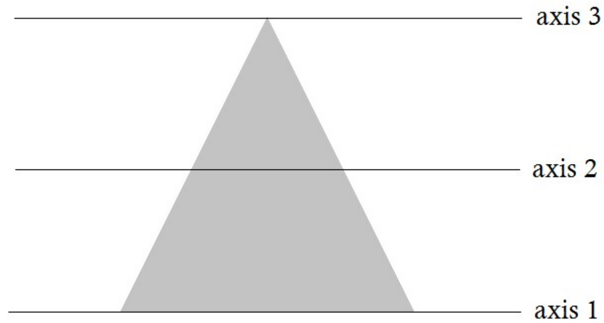
(II) the speed of the particle just before and just after the collision is equal if and only if the collision is perfectly elastic

Which of these options is correct?

(A) Both (I) and (II) are correct **(B)** (I) is correct and (II) is incorrect

(C) (II) is correct and (I) is incorrect **(D)** Both (I) and (II) are incorrect

5. Consider a thin uniform metallic sheet in the shape of an equilateral triangle. The three axes shown are parallel to each other, axis 1 being coincident with a side of the triangle, axis 3 passing through the opposite vertex and axis 2 being equidistant from the other two axes. If the moment of inertia of the triangular sheet about the three axes 1, 2 and 3 is I_1, I_2 and I_3 respectively, then:



(A) $I_3 > I_1 > I_2$ **(B)** $I_3 > I_2 > I_1$ **(C)** $I_1 > I_3 > I_2$ **(D)** $I_1 > I_2 > I_3$

6. A uniform rod of mass *M* and length *L* is pivoted about one of its ends such that it can rotate in the vertical plane. The rod is released from rest from the horizontal position. At the instant the rod is vertical, the angular momentum of the rod about its CM is:

(A) $M\left(\frac{gL^3}{3}\right)^{1/2}$ **(B)** $\frac{M}{4}\left(\frac{gL^3}{3}\right)^{1/2}$ **(C)** $\frac{M}{2}\left(\frac{gL^3}{6}\right)^{1/2}$ **(D)** $\frac{M}{2}\left(\frac{gL^3}{3}\right)^{1/2}$

7. Two identical masses travelling in the opposite direction with kinetic energy *K* and 9*K* have a head-on collision. If one of the masses comes to rest due to the collision, the kinetic energy of the other mass after the collision is:

(A) 3*K* **(B)** 4*K* **(C)** 5*K* **(D)** 9*K*

8. A hollow sphere of mass *M* and radius *R* rolls without slipping on a horizontal surface such that its angular velocity is ω. Consider the following statements:

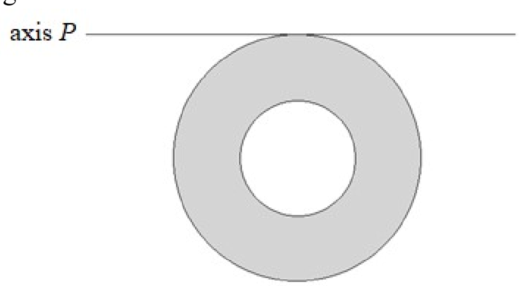
(I) The kinetic energy of the sphere is $\frac{4}{3}MR^2\omega^2$

(II) The angular momentum of the sphere about the point in contact with the surface is $\frac{5}{3}MR^2\omega^2$

Which of these options is correct?

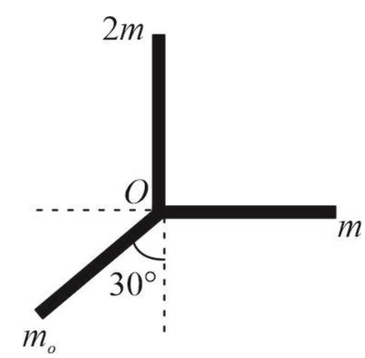
(A) Both (I) and (II) are correct **(B)** (I) is correct and (II) is incorrect
(C) (II) is correct and (I) is incorrect **(D)** Both (I) and (II) are incorrect

9. A concentric hole of radius $R/2$ is cut out of a uniform thin circular disc of radius R, resulting in the body shown. The mass of this resulting body is *M*. The moment of inertia of this body about an axis *P* lying in its plane and tangential to it is:



(A) $\frac{21}{16}MR^2$ **(B)** $\frac{19}{16}MR^2$ **(C)** $\frac{43}{32}MR^2$ **(D)** $\frac{39}{32}MR^2$

10. Three thin, uniform rods of the same length and mass m_0, $2m$ and m are joined to form a composite body with all three rods in the same plane, and this body is pivoted at the junction point O. The body remains in equilibrium with the rods of mass $2m$ and m being vertical and horizontal respectively, as shown. All three rods are in the vertical plane. The correct relation between m_0 and m is:

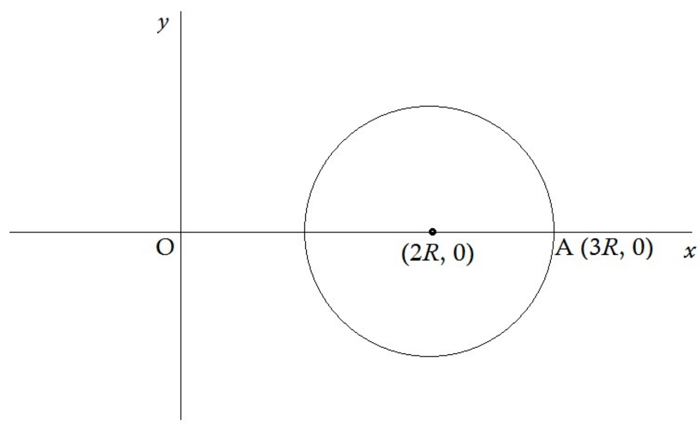


(A) $m_0 = \frac{\sqrt{3}}{2}m$ **(B)** $m_0 = 2m$ **(C)** $m_0 = \frac{2}{\sqrt{3}}m$ **(D)** $m_0 = \frac{m}{2}$

11. A uniform solid cylinder and a uniform solid sphere of radius R and $2R$ respectively are allowed to roll down two identical, fixed inclined planes, starting from rest. The friction between the objects and the inclined planes is sufficient to allow pure rolling. Let the time taken by the CMs of the objects to descend the same height h be t_1 and t_2 respectively. Then, $\frac{t_1}{t_2}$ is equal to:

(A) $\left(\frac{14}{15}\right)^{1/2}$ **(B)** $\left(\frac{15}{16}\right)^{1/2}$ **(C)** $\left(\frac{16}{15}\right)^{1/2}$ **(D)** $\left(\frac{15}{14}\right)^{1/2}$

12. A particle of mass m moves on a circle of radius R centred at the point $(2R,0)$ in the anticlockwise direction with a constant speed v. At $t=0$, the particle is at the point $A(3R,0)$. At $t = \frac{\pi R}{4v}$, the magnitude of the angular momentum of the particle about the origin O is:



(A) $\left(\sqrt{2}+1\right)mvR$ **(B)** $\left(\sqrt{2}-1\right)mvR$ **(C)** $\left(1+\frac{1}{\sqrt{2}}\right)mvR$ **(D)** $\left(1-\frac{1}{\sqrt{2}}\right)mvR$

13. Two small disks A and B of mass m and $2m$ are moving on a smooth horizontal table with velocity $\vec{v}_A = 4\,\hat{i}$ and $\vec{v}_B = \hat{i} - 3\,\hat{j}$ (The surface of the table is the X-Y plane). At $t=0$, the disks collide at the origin. At $t = 2\ s$, the coordinates of the centre of mass of the two disks are:

(A) $(2,-2)$ **(B)** $(4,-2)$ **(C)** $(4,-4)$ **(D)** $(2,-4)$

14. A particle initially at rest splits into three fragments of equal mass. Immediately after the splitting, two of the fragments are moving in mutually perpendicular directions with the same speed v. The speed of the third fragment immediately after the splitting is:

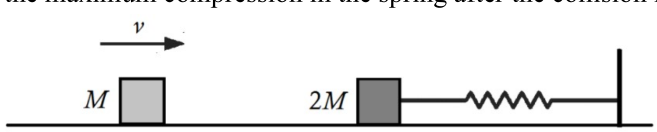
(A) $\frac{v}{\sqrt{2}}$ **(B)** v **(C)** $\sqrt{2}\,v$ **(D)** $2v$

15. A uniform rod of length L and mass M is pivoted horizontally at two points A and B. The point A is one of the ends of the rod and the point B is at a distance $L/4$ from A. Assuming that the forces acting on the rod from the pivots at A and B are directed vertically, these forces are respectively:

(A) mg downwards, $2mg$ upwards **(B)** mg upwards, $2mg$ downwards

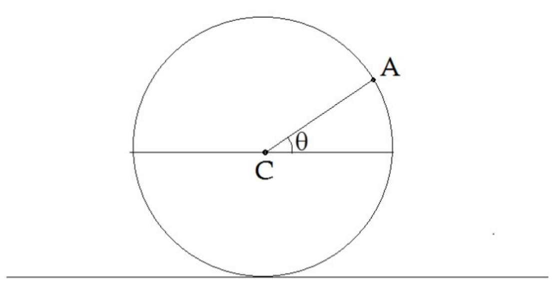
(C) $\frac{mg}{2}$ downwards, $\frac{3mg}{2}$ upwards **(D)** $\frac{mg}{2}$ upwards, $\frac{3mg}{2}$ downwards

16. A block of mass $2M$ is connected to an ideal spring of spring constant k and placed on a smooth horizontal surface. The other end of the spring is connected to a fixed support. The block is initially at rest. Another block, of mass M, moving with velocity v collides with the block. If the collision is perfectly inelastic, the maximum compression in the spring after the collision is:



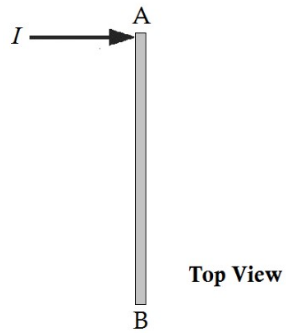
(A) $\sqrt{\frac{2M}{k}}\,v$ **(B)** $\sqrt{\frac{M}{2k}}\,v$ **(C)** $\sqrt{\frac{M}{3k}}\,v$ **(D)** $\sqrt{\frac{M}{6k}}\,v$

17. A uniform disc rolls without slipping on a horizontal surface such that the velocity of its centre is v, directed towards right in the figure shown. If the line AC makes angle $\theta = 30^o$ with the horizontal, the instantaneous speed of the point A on the rim of the disc is:



(A) $2v$ **(B)** $\sqrt{3}\,v$ **(C)** $\frac{\sqrt{5}}{2}v$ **(D)** v

18. A thin uniform rod AB of mass M is at rest on a smooth horizontal surface. An impulse I is given to the rod at its end A, in a direction perpendicular to the rod. The velocity of the end A immediately afterwards is:

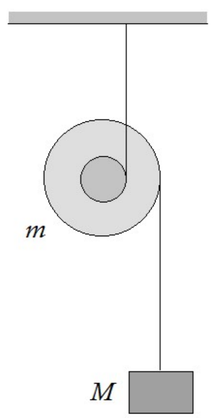


(A) $\frac{I}{M}$ **(B)** $\frac{2I}{M}$ **(C)** $\frac{3I}{M}$ **(D)** $\frac{4I}{M}$

19. The potential energy of a particle is given by: $U(x) = 16 - \frac{1}{4}(x-1)(x-5)$. Which of these options is correct?

(A) $x = 3$ is a stable equilibrium
(B) $x = 3$ is an unstable equilibrium
(C) $x = 6$ is a stable equilibrium
(D) $x = 6$ is an unstable equilibrium

20. A double pulley of smaller radius r and larger radius R, and mass m, is arranged with two massless strings and a block of mass M as shown. Both strings are wound on the pulley. The system remains in equilibrium if:



(A) $\frac{M}{m} = \frac{R}{R-r}$ **(B)** $\frac{M}{m} = \frac{r}{R-r}$ **(C)** $\frac{M}{m} = \frac{R+r}{R-r}$ **(D)** $\frac{M}{m} = \frac{R-r}{R+r}$

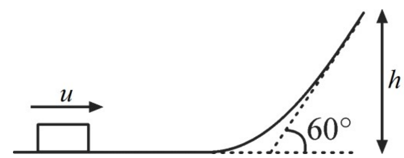
SPACE FOR ROUGH WORK

SECTION-2

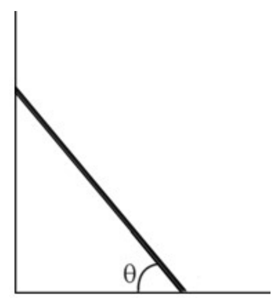
This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. A particle of mass 1 kg is initially at rest. At $t = 0$, a time-varying force $F = (8 - 3t)$ starts acting on the particle (t is time in seconds, and the force F is in Newton). The power (in Watts) being delivered by the force to the particle at $t = 2\ s$ is ________.

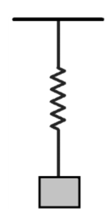
22. A block is travelling with velocity $u = 4\sqrt{2}\ m/s$ on a smooth horizontal surface towards a fixed smooth track of height h = 0.8 m that makes an angle 60^o with the horizontal near its top. The velocity (in m/s) of the block at the highest point in its path is _______. $\left(g = 10\ m/s^2\right)$



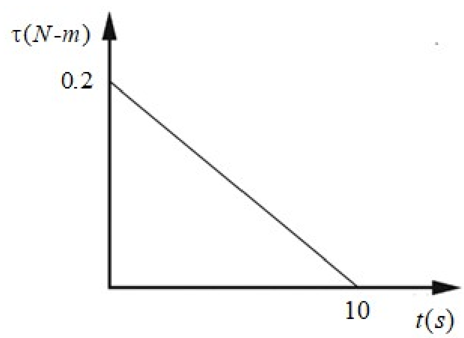
23. A thin rod of mass 4 kg rests in equilibrium against a smooth vertical wall, making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with the rough horizontal ground as shown. The force (in Newton) exerted by the rod on the wall is __________. $\left(g = 10\ m/s^2\right)$



24. A block of mass 1 kg is suspended vertically by an ideal spring of spring constant 300 N/m as shown, and the upper end of the spring is fixed. The block is released from rest when the spring is in its natural length. If the velocity of the block after it has moved down by a distance 5 cm is X cm/s, then X is ___________. $\left(g = 10\ m/s^2\right)$



25. A solid cylinder of mass 2 kg and diameter 40 cm mounted to rotate about its central axis is initially at rest. A torque begins to act on the cylinder at $t = 0$. The variation with time of this torque is given in the graph. The angular speed (in rad/s) of the cylinder at $t = 10$ s is ____________.



SPACE FOR ROUGH WORK

PART - II : CHEMISTRY — 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The initial pressure of $COCl_2$ is 1000 torr. The total pressure of the system becomes 1500 torr, when the equilibrium $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ is attained at constant temperature. The value of K_p of a reaction is:

(A) 1500 Torr **(B)** 1000 Torr **(C)** 2500 Torr **(D)** 500 Torr

2. In a chemical equilibrium the rate constant of the backward reaction is 7.5×10^{-4} and the equilibrium constant is 1.5. So the rate constant of the forward reaction is:

(A) 2×10^{-3} **(B)** 15×10^{-4} **(C)** 1.125×10^{-3} **(D)** 9.0×10^{-4}

3. A monoprotic acid in a 0.1 M solution ionizes to 0.001%. Its ionization constant is:

(A) 1.0×10^{-3} **(B)** 1.0×10^{-6} **(C)** 1.0×10^{-8} **(D)** 1.0×10^{-11}

4. When 0.1 mole of ammonia is dissolved in sufficient water to make 1 litre of solution. The solution is found to have a hydroxide ion concentration of 1.34×10^{-3}. The dissociation constant of ammonia is:

(A) 1.8×10^{-5} **(B)** 1.6×10^{-6} **(C)** 1.34×10^{-3} **(D)** 1.8×10^{-4}

5. 100 c.c. of N/10 NaOH solution is mixed with 100 c.c. of N/5 HCl solution and the whole volume is made to 1 litre. The pH of the resulting solution will be:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 12

6. The addition of solid sodium carbonate to pure water causes.

(A) An increase in the hydronium ion concentration
(B) An increase in pH
(C) No change in pH
(D) A decrease in the hydroxide ion concentration

7. Which will undergo cationic hydrolysis?

(A) NaCl **(B)** CH_3COONa **(C)** $(NH_4)_2SO_4$ **(D)** $NaNO_3$

8. Which of the following solutions will have pH close to 1?

(A) $100ml, \frac{M}{5} HCl + 100ml, \frac{M}{5} NaOH$ **(B)** $55ml, \frac{M}{10} HCl + 45ml, \frac{M}{10} NaOH$

(C) $10ml, \frac{M}{10} HCl + 90ml, \frac{M}{10} NaOH$ **(D)** $75ml, \frac{M}{5} HCl + 25ml, \frac{M}{5} NaOH$

9. Two samples of CH_3COOH each of 10 g were taken separately in two vessels containing water of 6 litre and 12 litre respectively at 27°C. The degree of dissociation of CH_3COOH will be:

(A) More in 12 litre vessel **(B)** More in 6 litre vessel
(C) Equal in both vessels **(D)** half in 6 litre vessel than in 12 litre vessel

10. A 20 litre container at 400 K contains $CO_2(g)$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be:

(Given that: $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g), K_p = 1.6\text{ atm}$)

(A) 5 litre **(B)** 10 litre **(C)** 4 litre **(D)** 2 litre

11. For the reversible reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{heat}$. The equilibrium shifts in forward direction.

(A) By increasing the concentration of $NH_3(g)$

(B) By decreasing the pressure

(C) By decreasing the concentrations of $N_2(g)$ and $H_2(g)$

(D) By increasing pressure and decreasing temperature

12. What is $[H^+]$ in mol/L of a solution that is 0.20 M CH_3COONa and 0.10 M CH_3COOH? (K_a for $CH_3COOH = 2 \times 10^{-5}$)

(A) 3.5×10^{-4} **(B)** 1×10^{-5} **(C)** 1.8×10^{-5} **(D)** 9.0×10^{-6}

13. Which one of the following is not a conjugate acid-base pair?

(A) $HONO, NO_2^-$ **(B)** $CH_3NH_3^+, CH_3NH_2$

(C) $C_6H_5-COOH, C_6H_5COO^-$ **(D)** H_3O^+, OH^-

14. Iron fillings and water were placed in a 5 L vessel and sealed. The tank was heated to 1000°C. Upon analysis, the tank was found to contain 1.2 g of $H_2(g)$ and 54.0 g of $H_2O(g)$. If the reaction is represented as: $3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$, the value of equilibrium constant is:

(Mass number.: H = 1, O = 16)

(A) 0.2 **(B)** 0.04 **(C)** 0.008 **(D)** 0.0016

15. For the equilibrium: $LiCl \cdot 3NH_3(s) \rightleftharpoons LiCl \cdot NH_3(s) + 2NH_3(g);\ K_P = 9\text{ atm}^2$ at 27°C. A 8.21 L vessel contains 0.1 mole of $LiCl.NH_3(s)$. How many moles of $NH_3(g)$ should be added to the flask at this temperature to drive the backward reaction for completion? $\left(R = 0.0821\dfrac{\text{atm}-\text{L}}{\text{mol}-\text{K}}\right)$

(A) 0.8 **(B)** 1.0 **(C)** 1.2 **(D)** 1.1

16. The equilibrium constant for the mutarotation, $\alpha-D$–glucose $\rightleftharpoons$ $\beta-D$–glucose is 1.8. What percent of the $\alpha-$form remains under equilibrium?

(A) 35.7 **(B)** 64.3 **(C)** 55.6 **(D)** 44.4

17. For the reaction:

$2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$, the ratio $\frac{K_P}{P}$, where P is the total pressure of gases at equilibrium and $P_{Br_2} = \frac{P}{9}$ at a certain temperature, is:

(A) $\frac{1}{9}$ **(B)** $\frac{1}{81}$ **(C)** $\frac{1}{27}$ **(D)** $\frac{1}{3}$

18. For the gas phase reaction: $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$; $\Delta H = -43.5$ kcal. Which one of the following is true for the reaction: $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$? (Here K is equilibrium constant and T is temperature)

(A) K is independent of T **(B)** K decreases as T decreases
(C) K increases as T decreases **(D)** K varies with addition of NO

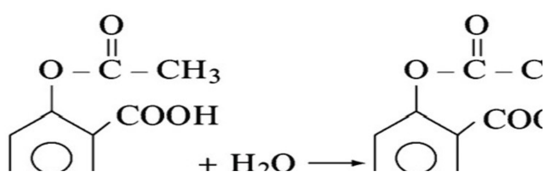
19. $XeF_6 + H_2O \rightleftharpoons XeOF_4 + 2HF$; Equilibrium constant $= K_1$.

$XeO_4 + XeF_6 \rightleftharpoons XeOF_4 + XeO_3F_2$; equilibrium constant $= K_2$

Then equilibrium constant for the following reaction will be: $XeO_4 + 2HF \rightleftharpoons XeO_3F_2 + H_2O$

(A) $\frac{K_1}{K_2}$ **(B)** $K_1 + K_2$ **(C)** $\frac{K_2}{K_1}$ **(D)** $K_2 - K_1$

20. The active ingredient in aspirin is acetyl salicyclic acid



With $K_a = 4.0 \times 10^{-9}$. The pH of the solution obtained by dissolving two aspirin tablets (containing 0.36 g of acetyl salicylic acid in each tablet) in 250 ml of water is (log 2 = 0.3)

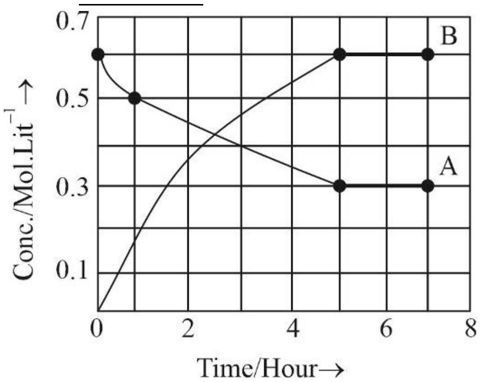
(A) 5.1 **(B)** 8.9 **(C)** 10.2 **(D)** 5.25

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. The progress of the reaction $A \rightleftharpoons nB$, with time is represented by the graph given below. The value of n is _________ .



22. In a neutral buffer solution of H_2CO_3 and HCO_3^- salt, the ratio $\frac{[HCO_3^-]}{[H_2CO_3]}$ is __________ .

(K_{a_1} of $H_2CO_3 = 4 \times 10^{-7}$; K_{a_2} of $H_2CO_3 = 5 \times 10^{-11}$)

23. What volume (in L) of water must be added to 1 L of 0.1 M solution of B (weak organic monoacidic base, ionization constant $= 10^{-5}$) to triple the % ionization of base?

24. What is pOH of an aqueous solution with $[H^+] = 10^{-2}$ M and $K_w = 2 \times 10^{-12}$ at a certain temperature? Report your answer after diving by 2 and round it off to the nearest whole number. (Given: log 2 = 0.3)

25. For equilibrium $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ the observed vapour density of N_2O_4 is 40 at 350 K. If percentage dissociation of $N_2O_4(g)$ at 350 K is x then $\frac{x}{5}$ is ________ .

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If $|z_1| = |z_2| = ... = |z_n| = 1$, then $\left|\dfrac{z_1 + z_2 + ... + z_n}{z_1^{-1} + z_2^{-1} + ... + z_n^{-1}}\right|$ is equal to :

(A) $\dfrac{1}{n}$ **(B)** n **(C)** 1 **(D)** $|z_1 + z_2 + ... + z_n|$

2. Point P lies on the line $l\{(x, y) \mid 3x + 5y = 15\}$. If '$P$' is also equidistant from the coordinate axes, then P can be located in which of the four quadrants ?

(A) Only I **(B)** Only II **(C)** I or II **(D)** Only IV

3. Centre of circles of radius 5 touching the line $3x + 4y = 7$ at (1, 1), is :

(A) $(4, 5)$ and $(-2, -3)$ **(B)** $(-4, -5)$ and $(2, 3)$

(C) $(-4, -5)$ and $(6, 7)$ **(D)** None of these

4. In the expansion of $(5^{1/2} + 7^{1/8})^{1024}$, the number of integral terms is :

(A) 128 **(B)** 129 **(C)** 130 **(D)** 131

5. The ratio of the coefficient of x^{10} in $(1 - x^2)^{10}$ and the term independent of x in $\left(x - \dfrac{2}{x}\right)^{10}$, is :

(A) 1 : 16 **(B)** 1 : 32 **(C)** 1 : 64 **(D)** None of these

6. Let z, w be two complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg(zw) = \pi$. Then, $\arg(z)$ equals :

(A) $\dfrac{\pi}{4}$ **(B)** $\dfrac{\pi}{2}$ **(C)** $\dfrac{3\pi}{4}$ **(D)** $\dfrac{5\pi}{4}$

7. The number of integral points (integral point means both the coordinates should be integer) exactly in the interior of the triangle with the vertices (0, 0), (0, 21) and (21, 0) is :

(A) 133 **(B)** 190 **(C)** 233 **(D)** 105

8. The angle at which the circles $(x-1)^2 + y^2 = 10$ and $x^2 + (y-2)^2 = 5$ intersects is :

(A) $\dfrac{\pi}{6}$ **(B)** $\dfrac{\pi}{4}$ **(C)** $\dfrac{\pi}{3}$ **(D)** $\dfrac{\pi}{2}$

9. Given that the family of lines, $a(3x + 4y + 6) + b(x + y + 2) = 0$. The line of the family situated at the greatest distance from the point $P(2, 3)$ has equation :

(A) $4x + 3y + 8 = 0$ **(B)** $5x + 3y + 10 = 0$

(C) $15x + 8y + 30 = 0$ **(D)** None of these

10. The number of terms in the expansion of $\left(x^2 + 1 + \dfrac{1}{x^2}\right)^n$, $n \in N$ is :

(A) $2n$ **(B)** $n + 1$ **(C)** $(2n + 1)$ **(D)** $3n + 1$

11. If in the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the seventh term from the beginning to the seventh term from end is equal to 1/6, then n is equal to :

(A) 3 **(B)** 6 **(C)** 9 **(D)** None of these

12. If in the expansion of $(1+x)^m(1-x)^n$, the coefficients of x and x^2 and 3 and -6 respectively, then m is :

(A) 6 **(B)** 9 **(C)** 12 **(D)** 24

13. If $\alpha = e^{i(2\pi/n)}$, then $(11-\alpha)(11-\alpha^2)....(11-\alpha^{n-1})$ is equal to :

(A) 11^{n-1} **(B)** $\frac{11^n-1}{10}$ **(C)** $\frac{11^{n-1}-1}{10}$ **(D)** $\frac{11^{n-1}-1}{11}$

14. (1, 2) is vertex of a square whose one diagonal is along the X-axis. The equations of sides passing through the given vertex are :

(A) $2x-y=0,\ x+2y+5=0$ **(B)** $x-2y+3=0,\ 2x+y-4=0$

(C) $x-y+1=0,\ x+y-3=0$ **(D)** None of these

15. In a ΔABC, side AB has the equation $2x+3y=29$ and the side AC has the equation, $x+2y=6$. If the mid-point of BC is (5, 6), then the equation of BC is :

(A) $x-y=-1$ **(B)** $5x-2y=13$

(C) $21x+31y=291$ **(D)** $3x-4y=-9$

16. If the line $3x-4y-8=0$ divides the circumference of the circle with centre $(2, -3)$ in the ratio 1 : 2. Then, the radius of the circle is :

(A) 2 **(B)** $2\sqrt{2}$ **(C)** 4 **(D)** $4\sqrt{2}$

17. The remainder, when $(15^{23}+23^{23})$ is divided by 19, is :

(A) 4 **(B)** 15 **(C)** 0 **(D)** 18

18. The sum of the coefficients of all the even powers of x in the expansion of $(2x^2-3x+1)^{11}$, is :

(A) $2\cdot 6^{10}$ **(B)** $3\cdot 6^{10}$ **(C)** 6^{11} **(D)** None of these

19. The sum of the series $\sum_{r=0}^{10} {}^{20}C_r$, is :

(A) 2^{20} **(B)** 2^{19} **(C)** $2^{19}+{}^{19}C_{10}$ **(D)** $2^{19}-\frac{1}{2}\,{}^{20}C_{10}$

20. If $(7+4\sqrt{3})^n = \alpha+\beta$, where α is a positive integer and β is a proper fraction, then :

(A) $(1-\beta)(\alpha+\beta)=1$ **(B)** $\beta(\alpha+\beta)=1$

(C) $(\alpha+\beta)(2-\beta)=1$ **(D)** $\beta(\alpha-\beta)=1$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. If $x = a + ib$ is a complex number such that $x^2 = 3 + 4i$ and $x^3 = 2 + 11i$, where $i = \sqrt{-1}$, then $(a+b)$ is equal to ________.

22. If the point $(1, \lambda)$ always remains in the interior of the triangle formed by lines $y = x$, $y = 0$ and $x + y = 4$, where $\lambda \in (a, b)$. Then $a + b$ is ________.

23. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is tangent to C_1 at P and C_2 at Q is k, then $\frac{k}{4}$ is ________.

24. Circles are drawn through (1, 1) touching the circle $x^2 + y^2 - 6x + 2y + 1 = 0$. If r_1 and r_2 are the radii of smallest and largest circles. Then, the value of $(r_1 + r_2)^2 - (r_2 - r_1)^2$ equals ________.

25. If $C_r \equiv {}^{25}C_r$ and $C_0 + 5 \cdot C_1 + 9 \cdot C_2 + \ldots + (101) \cdot C_{25} = 2^{25} \cdot k$, then k is equal to ________.

SPACE FOR ROUGH WORK

End of JEE Main – 6 | JEE - 2023 (Gen 1 & 2)